

SATOSHIFLOW

Organizer-compatible BTC/USD strategy submission

1. Objective and supplied dataset

The objective is a deterministic long/short BTC/USD trend strategy implemented through the challenge's original **BackTester** interface. The supplied repository dataset covers 2018-2022. The strategy is generic and is designed to run unchanged on evaluator-provided OHLCV data.

The run used **btc_18_22_1d.csv** (1,826 daily rows), spanning 2018-01-01 through 2022-12-31. SHA-256: 32814cd9f3b83dc79a0092c1f1c24bd988dad4c4b230dcd421401025ac660748.

2. Official organizer-framework results

Official organizer metric	Value
Initial / final capital	\$1,000.00 / \$5,463.61
Net profit	\$4,463.61
Net return	446.36%
Sharpe ratio	1.012
Maximum drawdown	21.37%
Win rate	55.17%
Completed trades	29
Framework brokerage deducted	\$136.70
Buy-and-hold benchmark	23.64%

These headline values come only from the unmodified organizer framework after calling **bt.get_trades(1000)** and **bt.get_statistics()**. They are not mixed with the independent research engine.

3. Fixed strategy

The fixed parameters are Donchian 30, ADX 14 with threshold 20, EMA 200, ATR 14, and a 2.5 x ATR stop. A long requires close above the prior Donchian high and EMA, ADX >= 20, and DI+ > DI-. A short uses the symmetric conditions. The high/low-based trailing stop tracks the most favorable high for a long or low for a short. An opposite confirmed breakout may reverse the position.

4. Indicator formulas

TR = $\max(H-L, |H-C_{prev}|, |L-C_{prev}|)$. **ATR** is Wilder's recursive average of TR. **+DMI/-DM** retain the larger positive directional high/low move. **DI** = $100 \times \text{Wilder}(\text{DM}) / \text{ATR}$. **DX** = $100 \times |DI+ - DI-| / (DI+ + DI-)$, and **ADX** is Wilder-smoothed DX. **EMA** = $\alpha \times \text{close} + (1-\alpha) \times \text{prior EMA}$, $\alpha = 2/(\text{period}+1)$. Donchian thresholds are $\text{high.shift(1).rolling(30).max()}$ and $\text{low.shift(1).rolling(30).min()}$.

5. Timing and signal convention

The organizer executes a signal at the signal row's close. To prevent same-row bias, the strategy forms a decision only after completed candle t and writes its executable signal to candle $t+1$. Thus decision t executes at $t+1$ close. Signal values follow the exact state convention: 0 hold, +/-1 open or close, and +/-2 reverse. A separate mechanical penultimate-row decision closes any open position on the final row; an unexecutable last-row economic decision is discarded.

6. Brokerage and accounting

Initial capital is exactly \$1,000 and the configured brokerage is 0.15%. The unchanged organizer **TradePair.pnl()** deducts 0.15% of absolute trade quantity once per completed trade. That framework behavior is reported without alteration. The secondary research engine is deliberately stricter: it charges 0.15% on entry notional and again on exit notional and marks open positions to market daily.

7. Organizer equity path

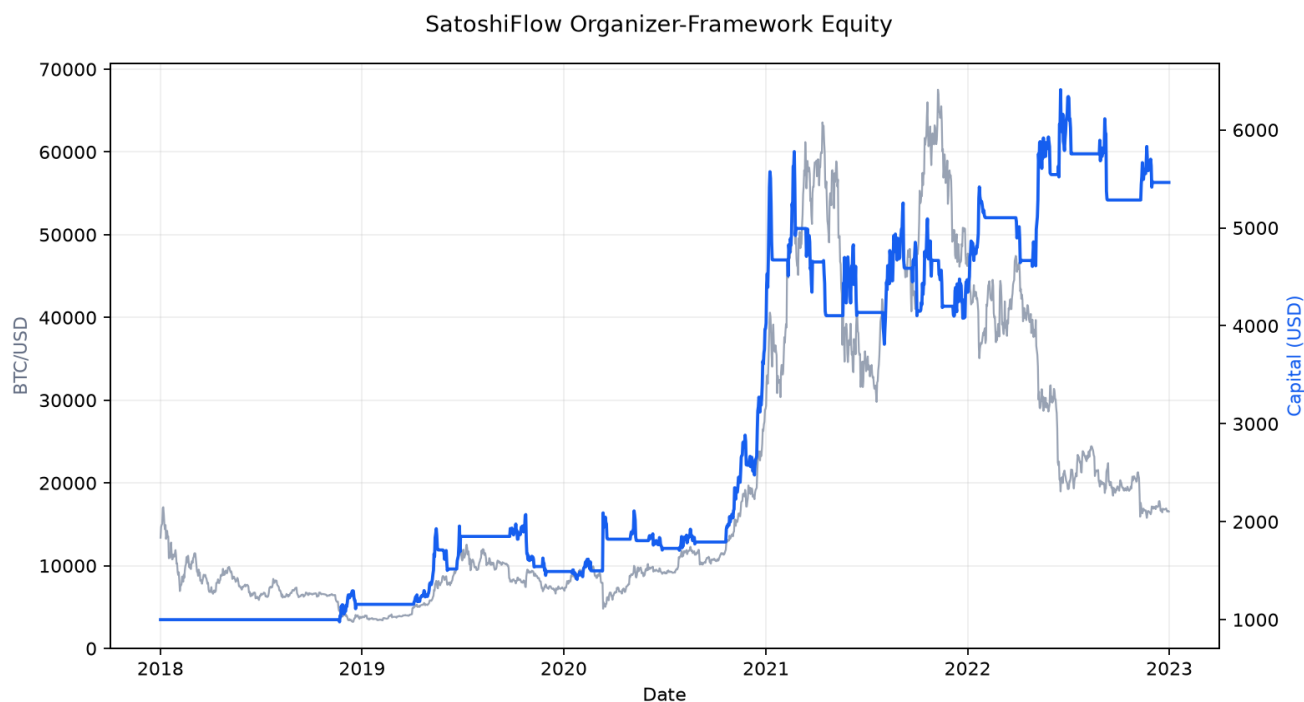


Figure 1. Organizer capital series with BTC close shown for context.

8. Integrity checks

LOOKAHEAD CHECK: PASS. Indicators and economic decisions match full-data history at five cutoffs. **SIGNAL SHIFT CHECK: PASS.** Every executable signal equals the prior row's decision. **SIGNAL VALIDITY: PASS.** A replay accepts every transition under organizer rules. **REPRODUCIBILITY CHECK: PASS.** Two complete organizer runs produce identical signals, trades, statistics, and capital.

9. Independent robustness check

The same fixed decisions were also evaluated with an independent next-open, daily mark-to-market simulator using two-sided fees. These values are secondary diagnostics and are intentionally separate from the organizer headline metrics.

Independent robustness metric	Value
Final equity	\$3,743.40
Net return	274.34%
Sharpe ratio	0.818
Mark-to-market drawdown	54.64%
Win rate	48.28%
Completed trades	29
Two-sided brokerage	\$237.02

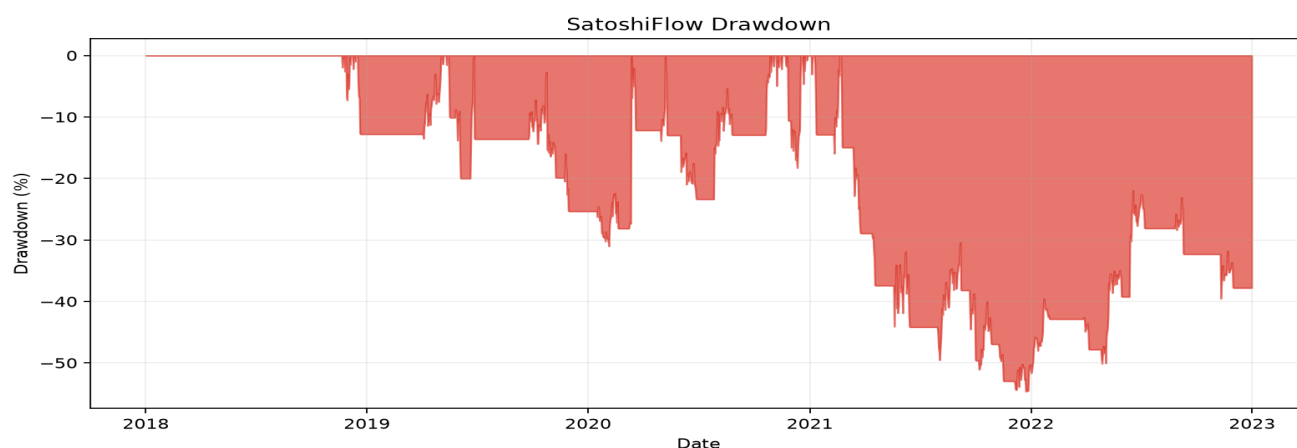


Figure 2. Independent simulator daily mark-to-market drawdown.

10. Limitations

The supplied data ends in 2022 and may differ from evaluator data in exchange, timezone, or candle construction. The organizer framework executes at next-bar close after signal shifting and uses its own one-deduction fee model; the robustness engine shows the sensitivity to next-open execution and two-sided fees. Neither engine models spread, slippage, funding, borrow constraints, latency, or market impact. Historical results do not guarantee live performance.

Source of truth: results/organizer/metrics.json. Every displayed headline value is loaded programmatically from that file.